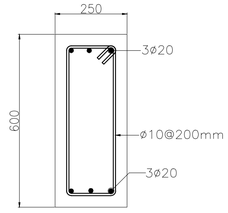
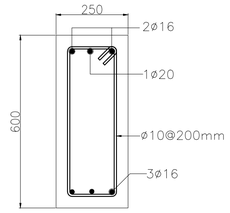
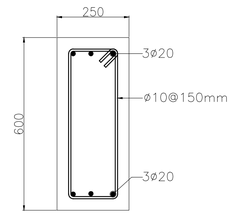
DRAWING NUMBER: MD-216-49BL-EG-CV-DED-1065
REV: D01



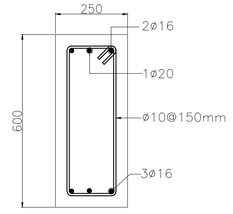
GRADE BEAM @ MIDSPAN GB1
SCALE 1:10 CAD-1063



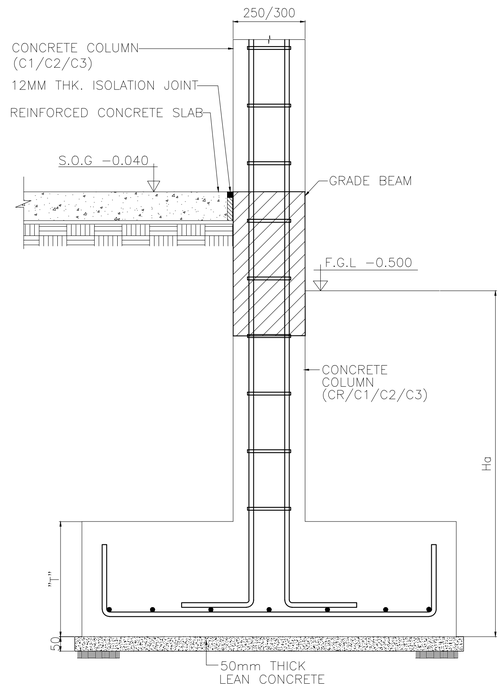
GRADE BEAM @ MIDSPAN GB2
SCALE 1:10 CAD-1063



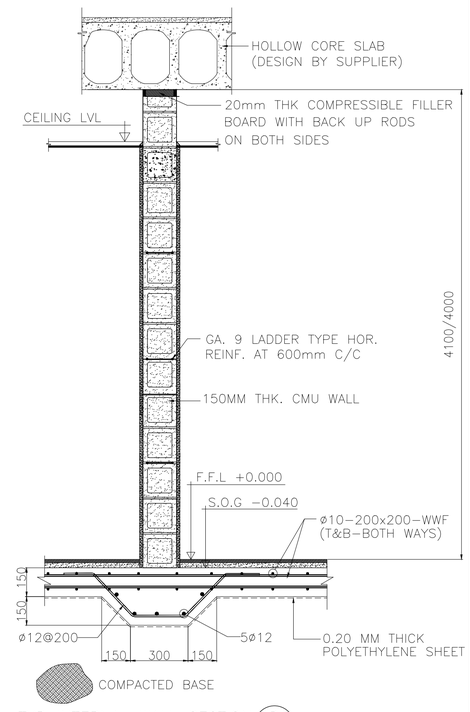
GRADE BEAM @ ENDSPAN GB1
SCALE 1:10 CAD-1063



GRADE BEAM @ ENDSPAN GB2
SCALE 1:10 CAD-1063



TYPICAL FOUNDATION SECTION
SCALE 1:10



TYP INTERNAL WALL SECTION D
SCALE 1:10 CAD-1064

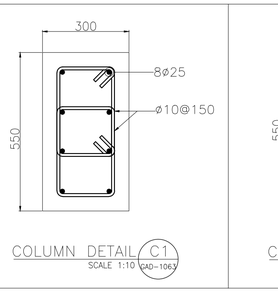
SCHEDULE OF FOOTINGS

FOOTING MARK	PLAIN CONCRETE			REINFORCED CONCRETE			BOTTOM REINFORCEMENT		TOP REINFORCEMENT		Ha	REMARKS
	L	W	T	L	W	T	SHORT BARS	LONG BARS	SHORT BARS	LONG BARS		
F1	2400	2400	50	2200	2200	400	ø12@150	ø12@150	–	–	1200	
F2	2400	2400	50	2200	2200	450	ø12@125	ø12@125	–	–	1200	
F3	1900	1900	50	1700	1700	400	ø12@150	ø12@150	–	–	1200	
CF1	3700	2400	50	3500	2200	400	ø12@125	ø12@125	ø12@250	ø12@250	1200	
CF2	3700	2400	50	3500	2200	450	ø12@125	ø12@125	ø12@250	ø12@250	1200	

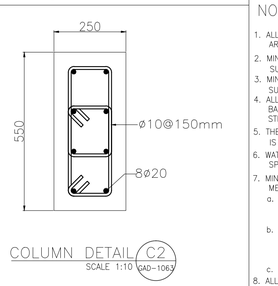
SCHEDULE OF GRADE BEAMS

BEAM MARK	DIMENSIONS		STEEL REINFORCEMENT				STIRRUPS
			BOTTOM		TOP		
	WIDTH	DEPTH	STRAIGHT	ADD.	STRAIGHT	ADD.	
GB1	250	600	3ø20	–	3ø20	–	PROVIDE ø10@150 IN THE ENDSPAN @DISTANCE OF L/3 FROM SUPPORT & ø10@200 IN THE MIDSPAN
GB2	250	600	3ø16	–	2ø16+1ø20	–	

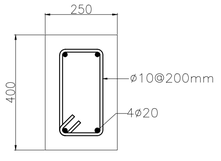
COLUMN DETAIL CR
SCALE 1:10 CAD-1063



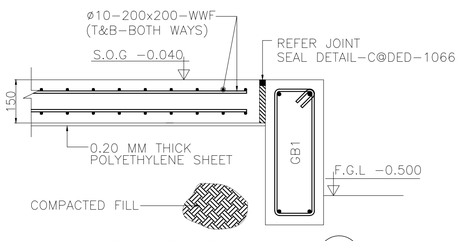
COLUMN DETAIL C1
SCALE 1:10 CAD-1063



COLUMN DETAIL C2
SCALE 1:10 CAD-1063



COLUMN DETAIL C3
SCALE 1:10 CAD-1063



TYP SECTION FOR GRADE SLAB A
SCALE 1:10 CAD-1063

NOTES

1. ALL DIMENSIONS ARE GIVEN IN MILLIMETRES AND ELEVATIONS ARE IN METRES UNLESS OTHERWISE NOTED.
2. MINIMUM REINFORCED CONCRETE GRADE TO BE USED FOR SUPERSTRUCTURE & SUBSTRUCTURES IS C35(5000 PSI).
3. MINIMUM LEAN CONCRETE GRADE TO BE USED FOR THE SUB-STRUCTURES IS C15(2175 PSI).
4. ALL REINFORCING STEEL SHALL BE DEFORMED BILLET STEEL BARS CONFORMING TO ASTM-A615, WITH MINIMUM YIELD STRENGTH OF FY420 MPA .
5. THE NET SOIL BEARING CAPACITY CONSIDERED IN FOUNDATION DESIGN IS 570 KPA(REFER DOC 216-3000-EG-CV-RPT-1008)
6. WATER PROOFING INSULATION SHALL BE AS PER PROJECT SPECIFICATIONS & ITS MATERIAL TO BE APPROVED BY SITE ENGINEER.
7. MINIMUM CONCRETE COVER AS PER SBC-304-2018(TABLE 20.6.1.3.1) MEASURED FROM REBAR SHALL BE
   a. CONCRETE CAST AGAINST & PERMANENTLY EXPOSED TO EARTH (INCLUDING FOUNDATIONS OVER SUB-SLAB) FORMED CONCRETE EXPOSED TO EARTH OR WEATHER.....50MM
   b. CONCRETE NOT EXPOSED TO WEATHER NOR IN CONTACT WITH GROUND
      BEAMS, COLUMNS.....40MM
      SLABS, WALLS, JOISTS.....20MM
   c. CONCRETE SLAB OVER 50MM SUB-SLAB.....20MM
8. ALL DRAWINGS MUST BE READ IN CONJUNCTION WITH RELEVANT ARCHITECTURAL AND ENGINEERING DRAWINGS RELATED TO THE SAME SHEET AND ANY ITEM ON IT.
9. F.F.L +0.00 MENTIONED IS EQUIVALENT TO RL 913.500

LEGENDS
F/FDN.–FOUNDATION
CR–COLUMN NECK
C/COL.–COLUMN
FFL–FINISH FLOOR LEVEL
FGL–FINISH GROUND LEVEL
TOS–TOP OF SCREED
WWF–WELDED WIRE FABRIC
HCS–HOLLOW CORE SLAB
GB–GRADE BEAM
IJ–ISOLATION JOINT
CJ–CONTROL JOINT
BW's–BOTH WAYS
S.O.G–SLAB ON GRADE
ARCH–ARCHITECTURAL
HOR.–HORIZONTAL
REINF.–REINFORCEMENT
LVL.–LEVEL
RL.–REDUCED LEVEL
P.–PEDESTAL
℄ –CENTRE LINE OF COLUMN

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-6000-EG-CV-STD-1002	HOLLOW CORE SLAB STANDARD	D01	08/09/2023	AS-BUILT	--	AVG	--	SJM	
MD-216-6000-EG-CV-STD-1001	REINFORCEMENT STANDARD	C02	13/01/2021	ISSUED FOR COSTRUCTION & REVISED AS MARKED △C02	--	AVG	--	SJM	
MD-216-49BL-EG-AR-GAD-3012	BUILDING ELEVATIONS & SECTIONS	C01	06/07/2020	ISSUED FOR COSTRUCTION	--	AVG	--	SJM	
MD-216-49BL-EG-AR-GAD-3011	BUILDING PLANS	A01	31/05/2020	ISSUED FOR APPROVAL	--	AVG	--	SJM	

DESIGNER: معهب الخليج للاستشارات الهندسية، العبر، المملكة العربية السعودية
GULF CONSULT ARCHITECTS AND ENGINEERS

DRAWN	MAH	CHECKED	MF	APPROVED	FK
DATE	27/05/2020	DATE	28/05/2020	DATE	29/05/2020

DRAWING NUMBER: LABORATORY BUILDING-ST-001
REVISION: 0

LARSEN & TOUBRO DRAWING NUMBER: 019045_300_C_D_24027_001
REVISION: 3

OUTOTEC DRAWING NUMBER: 900707__GAU01__AL01__CTB17__69001
DOCUMENT ID: OU100688039
REVISION: 3

LARSEN & TOUBRO
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DRAWN	--	--
CHECKED	AVG	30/05/2020
ENGINEER	--	--
APPROVED	SJM	31/05/2020
EMPLOYER		
SCALE	1:150	A1

MANSOURAH - MASSARAH GOLD PROJECT
DETAIL DRAWING
PROCESS PLANT : LABORATORY BUILDING
DETAIL OF FDN. COL. & GRADE BEAM

DRAWING NUMBER: MD-216-49BL-EG-CV-DED-1065
REV: D01